

Sales Data Analysis Report

1) Introduction

This project focuses on analyzing retail sales data to understand customer behaviour, product, performance and revenue patterns.

The goal is to extract meaningful insights that can help business make better decisions and improve overall sales.

2) Data Cleaning

- Removed missing values from the dataset
- Filtered invalid records such as negative quantity and zero unit price
- Handled inconsistent or incorrect data entries
- Ensured data consistency for accurate analysis

3) Feature Engineering

A new column was created:

$\text{Total_Sales} = \text{Quantity} \times \text{Unit Price}$

This column helps in calculating total revenue generated from each transaction.

The dataset was analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn.

The analysis includes:

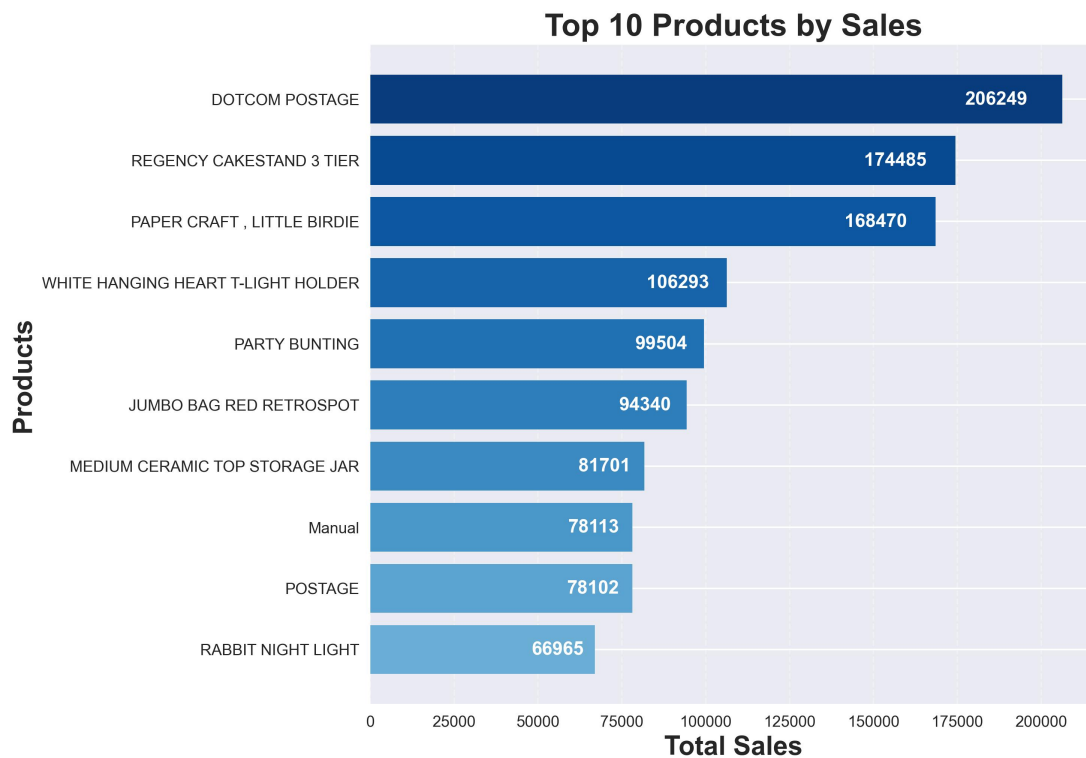
- Top products by total sales
- Top customers contributing to revenue
- Country-wise sales distribution

5) Key Insights

- A small number of customers contribute a large portion of total revenue
- Certain products generate significantly higher sales compared to others
- Sales are concentrated in specific countries
- Revenue distribution is uneven across customers and products
- Focusing on high-performing customers and products can improve overall business growth

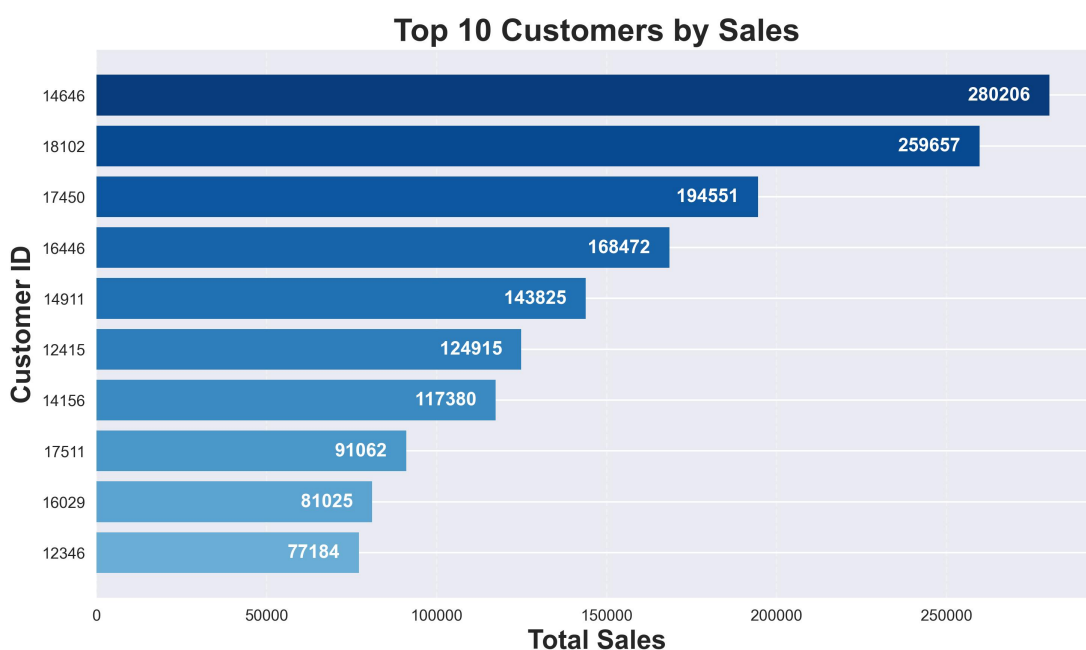
6) Visualizations

Top 10 Products by Sales



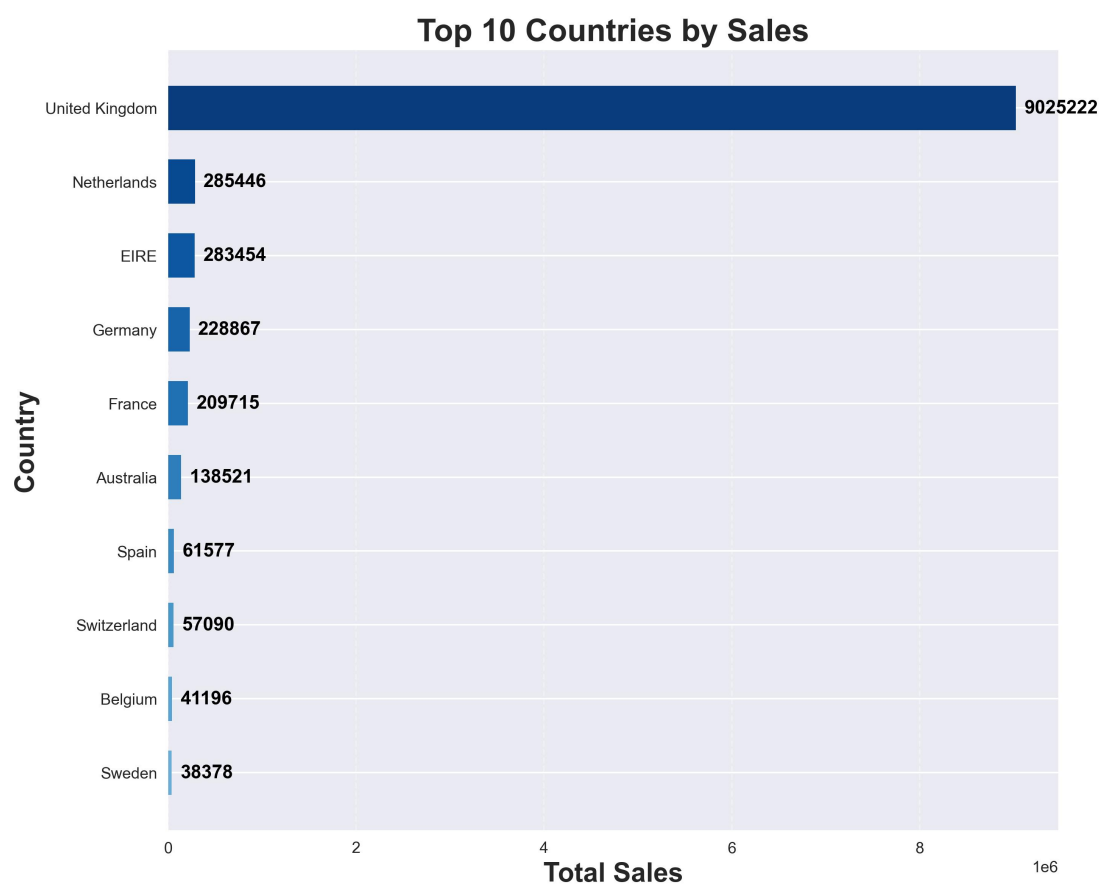
This chart highlights the top 10 products contributing the highest revenue, indicating that a small number of products generate a significant portion of total sales..

Top 10 Customers by Sales



This chart represents the top 10 customers based on total sales, showing that a small group of customers contributes disproportionately to overall revenue.

Top 10 Countries by Sales



This chart illustrates the top 10 countries by total sales, indicating that revenue is concentrated in a few regions.

7) Conclusion

This analysis provides clear insights into sales trends and customer behavior.

The results can help businesses focus on high-value customers and optimize product strategies to increase revenue.